Microsoft

Apache DevLake

A Decision-Maker’s Guide | Eldrick Wega · Microsoft

What this guide covers

Part 1 — Understanding DevLake. What it is, what you get, how the architecture works, and the core concepts.

Part 2 — Making the decision. When DevLake fits and when it doesn't, how it compares to alternatives, and what it costs to run.

Part 3 — Adopting DevLake. A phased path from laptop to production, plus the dashboards you'll actually see.

For engineering leaders, DevEx/platform teams, and architects evaluating Apache DevLake.

Part 1

Understanding DevLake

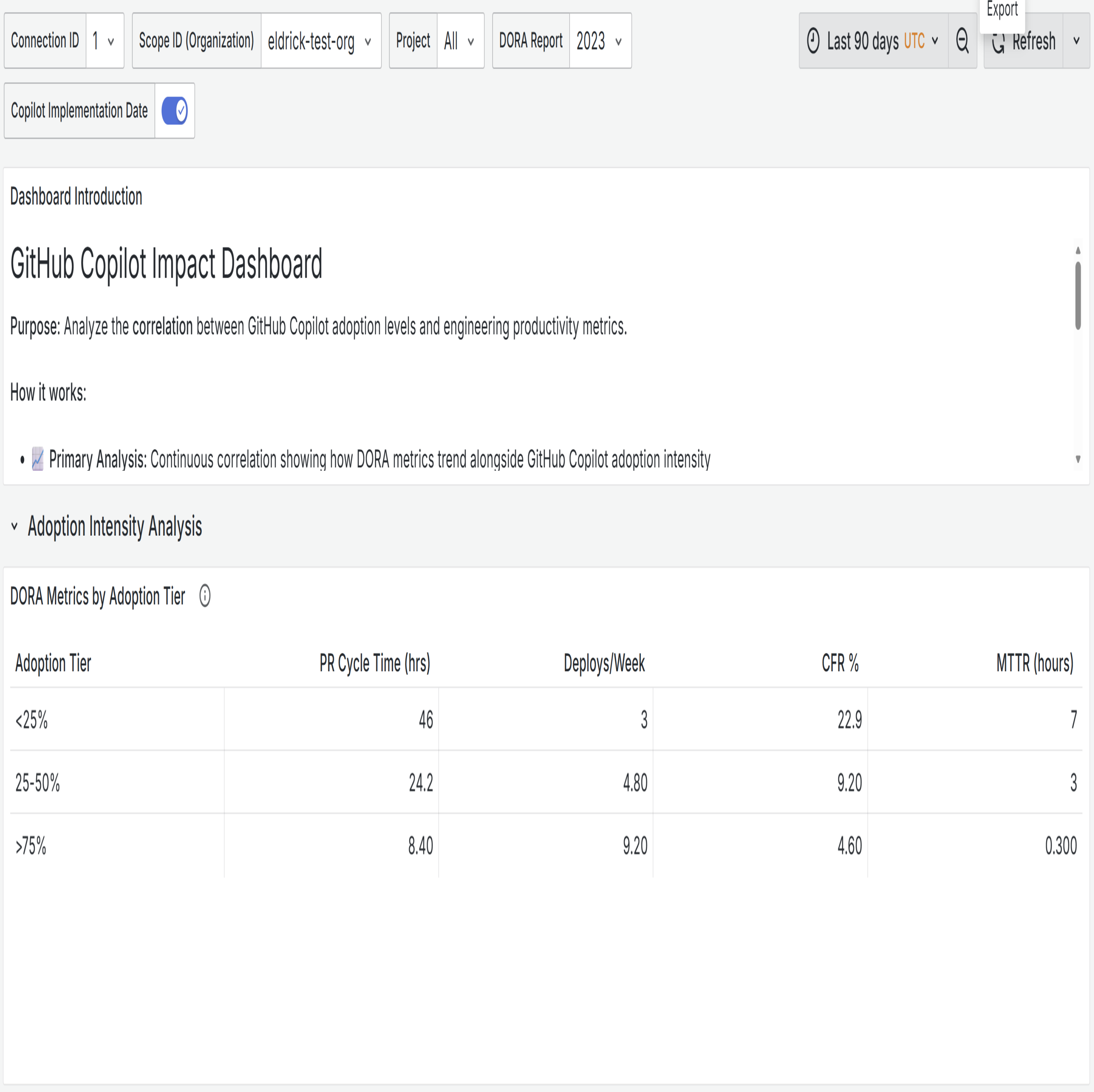
Copilot adoption tells you it's being used — not whether it's helping

GitHub's built-in Copilot metrics (seats, acceptance rates, editor mix) show rollout health. They don't tell you what that usage is doing to software delivery.

Answering the impact question requires correlating adoption with DORA metrics — which live in separate systems (CI/CD, issue tracker, source control).

Apache DevLake brings those signals into one place so the correlation becomes visible.

Example: teams in the >75% adoption tier show faster PR cycles, more frequent deploys, lower CFR, and faster recovery.



What Apache DevLake is

Open source. An Apache Software Foundation (Incubating) project — not a vendor product.

A dev data warehouse, not a dashboard. It ingests raw data from your DevOps tools into a normalised schema, then visualises it through Grafana.

20+ connectors. GitHub, Azure DevOps, Jira, Jenkins, GitLab, Bitbucket, SonarQube, PagerDuty, CircleCI, Copilot, and more.

Fully SQL-queryable. Every panel is backed by a SQL query you can inspect, modify, or extend.

What's included out of the box

DORA dashboards

- Deployment Frequency · Lead Time for Changes · Change Failure Rate · Mean Time to Restore.
- Built-in Elite/High/Medium/Low benchmarking against DORA research.

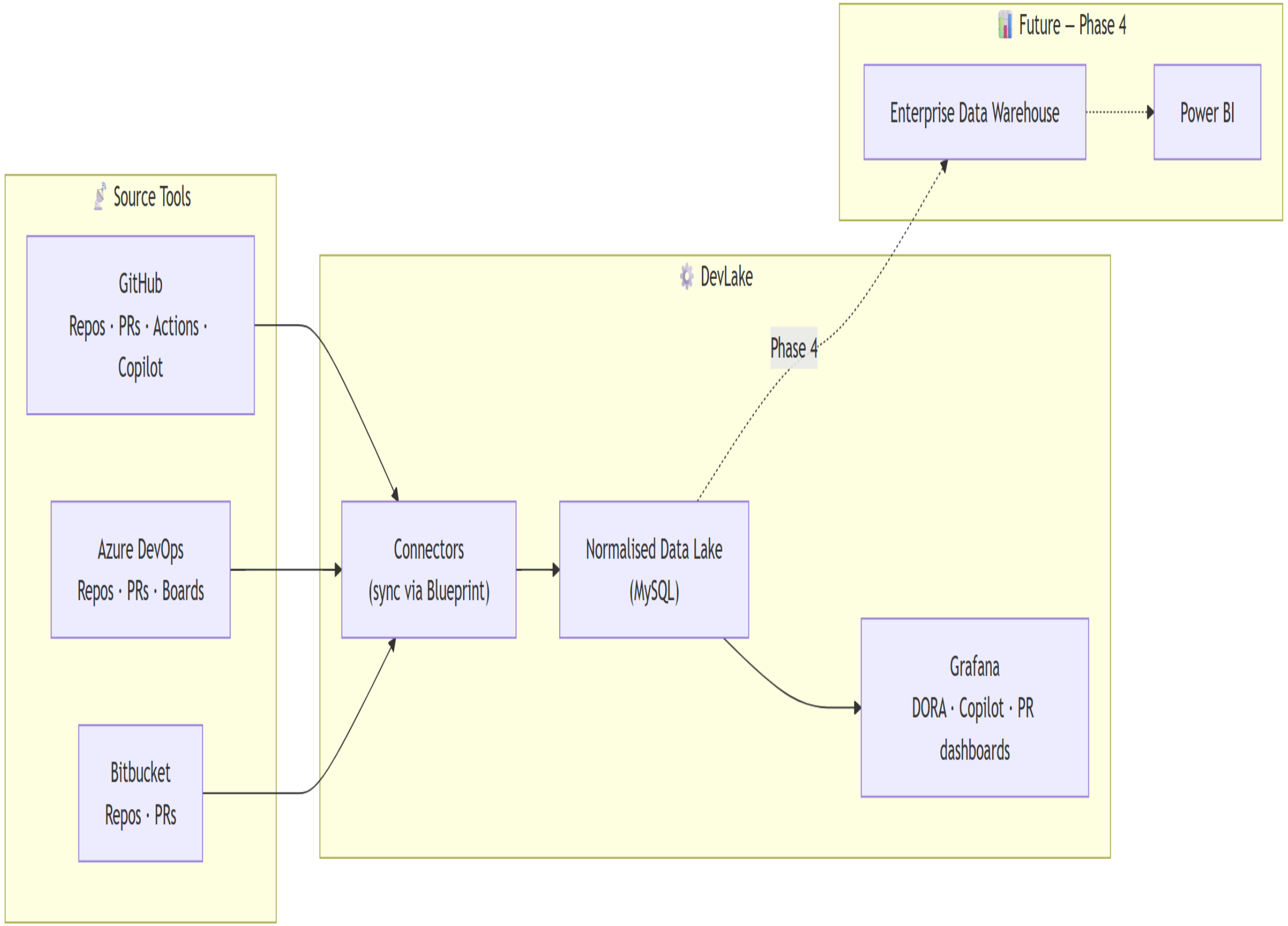
Copilot Adoption

- 30+ panels — DAU/WAU/MAU, acceptance rates, language/editor/model breakdowns, seat utilisation, feature mix (completions vs chat vs PR summaries).

Copilot Impact (Microsoft-built)

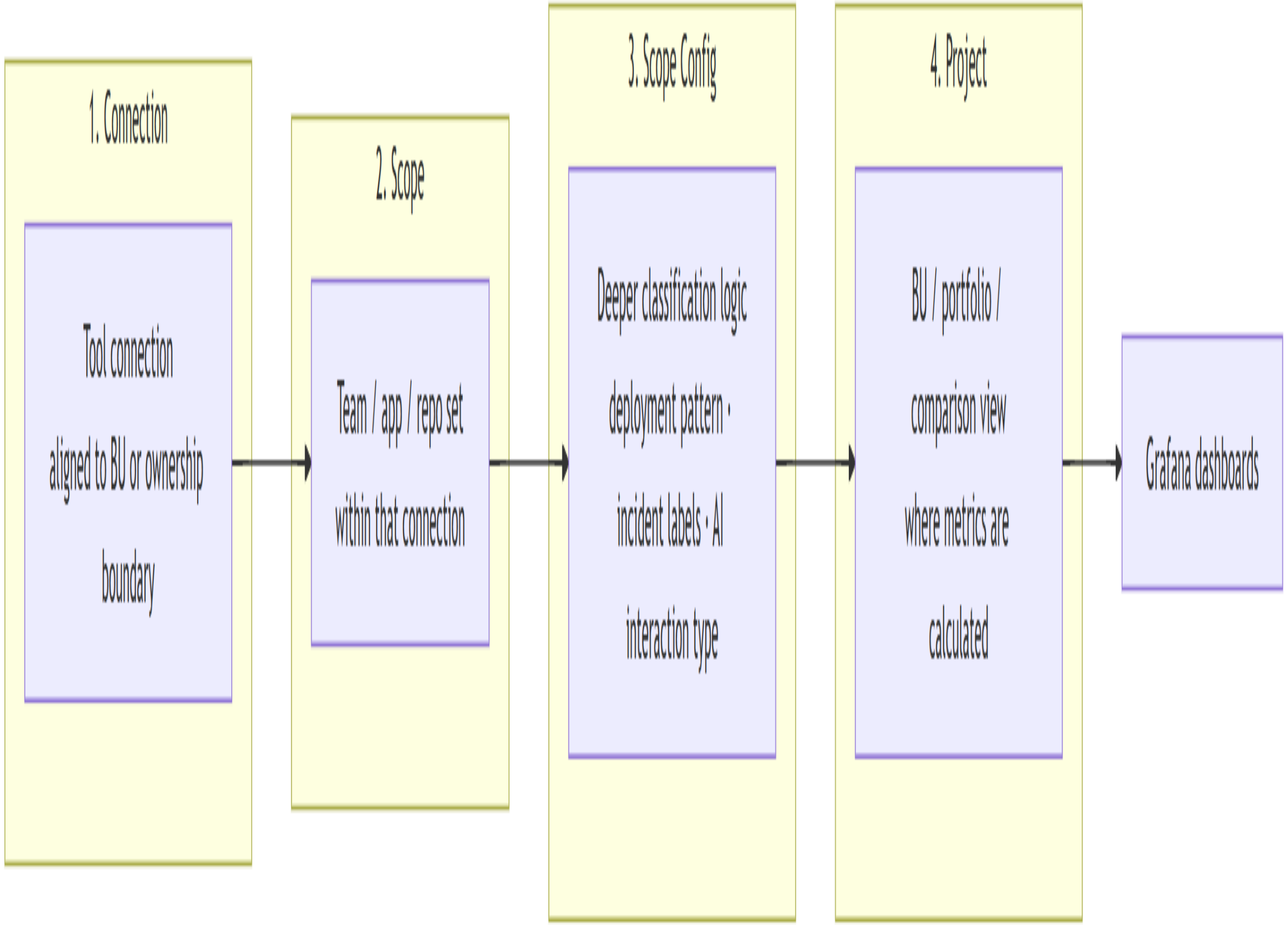
- Correlates adoption intensity (<25%, 25–50%, 50–75%, >75%) with DORA: PR cycle time, deployment frequency, CFR, MTTR, review time.

How it works



DevLake ingests data from your DevOps tools through connectors, normalises it into a MySQL data lake, and exposes it via Grafana dashboards.

Phase 4 (future): export to an enterprise data warehouse to blend engineering data with business/financial signals in Power BI.



Four concepts drive everything

Connection. Authenticated link to a tool (e.g., your GitHub org).

Scope. What to collect within a connection — a repo set, a Jira board.

Scope Config. Rules for how raw data maps to DORA concepts: deployment patterns, incident labels, environment naming.

Project. Logical grouping of scopes — DORA is calculated at this level.

Blueprint. Recurring sync schedule (created automatically).

Part 2

Making the decision

When DevLake fits — and when it doesn’t

Fits well when…

- You’ve rolled out Copilot and leadership wants to see measurable impact on delivery
- Your toolchain is mostly GitHub, Azure DevOps, Jira, Jenkins, GitLab, or Bitbucket
- You have a DevEx, platform, or engineering productivity team to operate it
- You want to own the data and extend with SQL
- You’re comfortable running self-hosted Grafana

Less suitable when…

- You’re a small team without dedicated DevEx ownership
- You have no CI/CD or incident signals (DORA depends on both)
- You need a fully managed SaaS with no operational overhead
- Your stack is primarily niche tools with no DevLake connector
- You want out-of-box insights with zero configuration

DevLake in context

Capability	Apache DevLake	GitHub built-ins	Commercial DevEx *(LinearB, Jellyfish, Swarmia)*
Copilot usage data	Yes (normalised)	Yes (native)	Via integrations
DORA metrics	Built-in	No	Yes
Cross-tool correlation	Yes (SQL)	No	Yes (proprietary)
SQL-level extensibility	Full	Limited	Limited
Cost model	OSS + infra (~$30–50/mo on Azure)	Included	$ per seat

Not a feature scorecard — a shape-of-landscape view. Choose based on ownership model, data access, and your team's extensibility needs.

Cost, ownership, and the fine print

Cost & operations

- Local: free (Docker on a developer laptop)
- Azure: ~$30–50 USD/month — Azure Container Instances + managed MySQL + Key Vault
- Owned by a DevEx, platform, or engineering productivity team — not individual devs
- Access via PATs or service credentials per data source

Things to know

- Apache Software Foundation *Incubating* — maturing project, not yet top-level
- Some tools need webhooks or custom plugins (e.g., Azure Pipelines)
- Data freshness depends on Blueprint sync cadence (daily default)
- DORA accuracy depends on Scope Config (deployment patterns, incident labels) being set correctly

Part 3

Adopting DevLake

A practical adoption path

Phase 1 — Foundation

~1 week

Local Docker deployment. Connect one or two source tools. Onboard 2 pilot teams. Prove DORA + Copilot data flows end-to-end.

Phase 2 — Expand

weeks 2–4

Move to Azure-hosted instance. Add remaining connectors. Onboard priority teams or business units. Produce first cross-team AI impact views.

Phase 3 — Optimise

month 2

Tune Scope Configs. Build comparison dashboards across teams. Add PR lifecycle metrics. Migrate source hosts if in progress.

Phase 4 — Mature

month 3+

Extend with custom plugins or webhooks for unsupported tools. Integrate incidents. Export to enterprise data warehouse for business blending.

Install & initialise

```
gh extension install DevExpGBB/gh-devlake
gh devlake init
```

Getting started with gh-devlake

One command — install the GitHub CLI extension

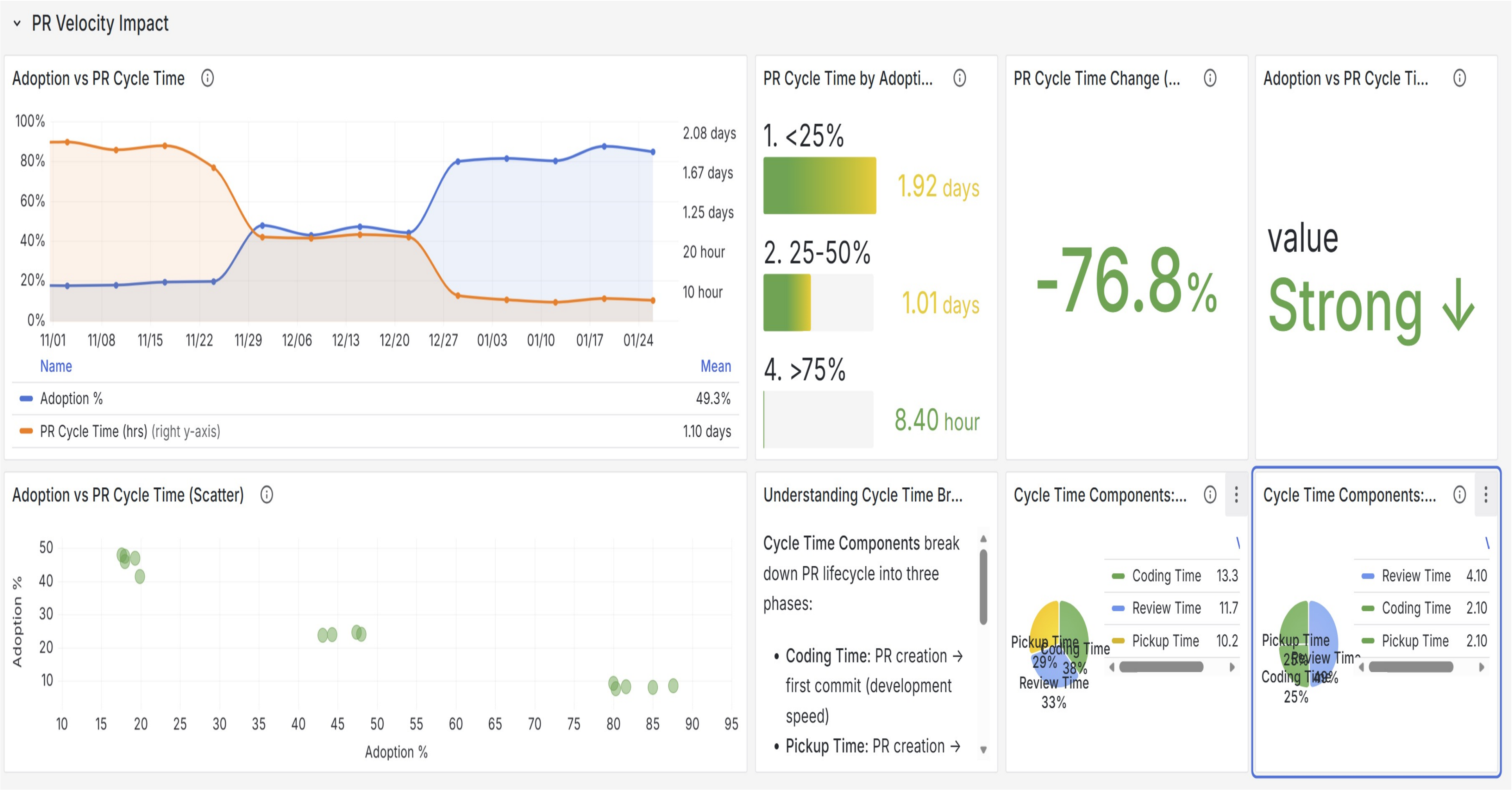
Deploy anywhere — local Docker or Azure via Bicep (~$30–50/mo)

Guided wizard — deploy → connect → scope → project → first sync

Dashboards live at http://localhost:3002 (admin / admin)

Repo: github.com/DevExpGbb/gh-devlake

Example: PR Velocity Impact



Correlating Copilot adoption with PR cycle time reveals a −76.8% change in cycle time between the lowest and highest adoption tiers. Every panel is a SQL query you can inspect and extend.

Resources & next steps

Read

- devlake.apache.org
- github.com/apache/incubator-devlake
- DORA & Copilot metric docs

Deploy

- gh-devlake (GitHub CLI extension)
- Docker Compose (local)
- Azure Bicep templates

Reference

- DORA research: dora.dev
- Blueprint & Scope Config guide
- Connector catalogue (20+ tools)

Dashboards

- Grafana at localhost:3002
- Default login: admin / admin
- DORA + AI Impact presets included

Community

- Apache incubator mailing list
- DevLake Slack workspace
- GitHub Discussions & Issues

Next steps

- Run a two-team local pilot
- Agree on DORA + AI metric set
- Identify DevEx/platform owner